Neurodevelopmental Disorder With Dysmorphic Facies And Variable Seizures (NEDDFAS): A Comprehensive Disease Report

MONDO: MONDO:0031011 · **OMIM:** #619264 · **Causal gene:** *EMC10* (HGNC:28974; chromosome 19q13.33) · **Inheritance:** Autosomal recessive · **Category:** Mendelian

Summary

Neurodevelopmental Disorder With Dysmorphic Facies And Variable Seizures (NEDDFAS) is an ultra-rare autosomal recessive syndromic intellectual-disability disorder caused by **biallelic loss-of-function (LoF) variants in *EMC10***, the gene encoding subunit 10 of the **endoplasmic reticulum (ER) membrane protein complex (EMC)**. The EMC is a highly conserved, membrane-embedded insertase that mediates the insertion and folding of **tail-anchored and multipass transmembrane proteins** into the ER membrane. NEDDFAS was first delineated in 2021 in 13 individuals from 7 consanguineous families who shared a recurrent homozygous frameshift variant, *EMC10* c.287delG (p.Gly96Alafs*9) (PMID: 33531666), and the molecular and clinical spectrum was expanded in 2022 and by subsequent case reports.

The disorder presents in **infancy or early childhood** with **global developmental delay/intellectual disability (GDD/ID) of variable severity**, **variable seizures**, **microcephaly**, **minor facial dysmorphism** (elongated/long face, curly hair), and **variable renal and skeletal features** (cubitus valgus, arachnodactyly, scoliosis), with additional reports of gingival hyperplasia. The intellectual disability is static (non-progressive) and lifelong. A single recurrent allele — **c.287delG (p.Gly96Alafs*9)** — accounts for the majority of reported patients; remarkably, it arose independently on two distinct founder haplotypes at a likely hypermutable site, explaining its recurrence across unrelated international, consanguineous families.

The proposed mechanism is that **biallelic EMC10 LoF reduces functional EMC insertase capacity**, impairing the biogenesis of membrane-protein clients (receptors, ion channels, transporters) in developing neurons and other tissues, leading to abnormal neurodevelopment and the multisystem phenotype. There is **no disease-specific or disease-modifying therapy**; management is supportive and symptomatic (antiseizure medication, early developmental intervention, rehabilitation, orthopedic and dental care) with **genetic counseling** for at-risk consanguineous/carrier families. Diagnosis is achieved almost exclusively through **whole-exome or whole-genome sequencing** with Sanger confirmation and segregation analysis.

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1. Disease Information

Overview. NEDDFAS is a Mendelian, autosomal recessive neurodevelopmental syndrome. It is defined molecularly by biallelic LoF variants in *EMC10* and clinically by the combination of developmental delay/intellectual disability, variable seizures, and variable dysmorphic facial features — as captured in its name. The first cohort described it as "a phenotypic spectrum of intellectual disability (ID) and global developmental delay (GDD), variable seizures and variable dysmorphic features" (PMID: 33531666).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0031011
OMIM	#619264
Causal gene	*EMC10* (HGNC:28974; NCBI Gene; located 19q13.33)
Reference transcript	NM_206538.3
Orphanet	Not separately catalogued at time of research (subsumed under rare syndromic intellectual disability)
ICD-11	Closest: syndromic intellectual developmental disorder (no NEDDFAS-specific code)
MeSH	No disease-specific descriptor; indexed under Neurodevelopmental Disorders / Intellectual Disability

Synonyms / alternative names. NEDDFAS; EMC10-related neurodevelopmental disorder; EMC10 deficiency; Neurodevelopmental disorder with dysmorphic facies and variable seizures.

Source of information. Evidence is derived from **individual patient reports and small aggregated case series** (~25–30 reported patients across fewer than 15 families), not from large disease-level registries or EHR-derived cohorts. All data come from the primary genetics literature.

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2. Etiology

Primary cause (genetic). NEDDFAS is a **monogenic Mendelian disorder** caused exclusively by **biallelic (homozygous or compound heterozygous) loss-of-function variants in *EMC10***. There are no known environmental, infectious, toxic, or multifactorial causes. The disease is fully genetic in origin.

Genetic risk factors. - **Causal variants:** biallelic *EMC10* LoF alleles. The recurrent frameshift **c.287delG (p.Gly96Alafs*9)** is the predominant pathogenic allele. Additional reported variants include the nonsense **c.343C>T (p.Arg115Ter)** (PMID: 40150819) and the frameshift **c.431del (p.Asp144AlafsTer3)** (PMID: 40741735), plus five additional novel biallelic LoF variants reported in the expanded cohort (PMID: 35684946). - **Consanguinity** is a major risk factor: most families are consanguineous, which raises the probability of homozygosity for rare recessive LoF alleles. - **Modifier genes:** none established. Variable expressivity exists but no specific genetic modifiers have been identified.

Environmental risk factors. None known or implicated. There are no toxins, exposures, lifestyle factors, or occupational hazards associated with disease occurrence or severity.

Protective factors. None known (genetic or environmental). As a fully penetrant recessive disorder, the only "protective" factor is the absence of a second pathogenic allele (i.e., heterozygous carriers are unaffected).

Gene–environment interactions. None described. The disorder is monogenic with no documented gene–environment modulation.

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3. Phenotypes

The NEDDFAS phenotype is **syndromic and multisystem**, with a neurodevelopmental core plus variable dysmorphic, renal, and skeletal features. The first cohort described "a phenotypic spectrum of intellectual disability (ID) and global developmental delay (GDD), variable seizures and variable dysmorphic features (elongated face, curly hair, cubitus valgus, and arachnodactyly)" (PMID: 33531666). The expanded cohort found ID "ranging from mild to severe" and "an overlap between the clinical features of EMC10- and EMC1-related disease" (PMID: 35684946). Forest-St-Onge et al. summarized it as "an autosomal recessive syndrome characterized by global developmental delay, intellectual disability, microcephaly, seizures, renal abnormalities, and minor facial dysmorphic features" (PMID: 40150819).

Phenotype	HPO term	Type	Frequency	Onset	Severity / course
Global developmental delay	HP:0001263	Clinical sign	Core (near-universal)	Infancy/ early childhood	Variable; static
Intellectual disability	HP:0001249	Clinical sign	Core (near-universal)	Early childhood	Mild→severe; non-progressive
Seizures	HP:0001250	Clinical sign	Variable	Childhood	Variable
Microcephaly	HP:0000252	Physical manifestation	Common	Congenital/ early	Variable
Facial dysmorphism (elongated/long face)	HP:0000276	Physical manifestation	Variable	Congenital	Minor
Curly hair	HP:0002212	Physical manifestation	Variable	Congenital	Minor
Cubitus valgus	HP:0002967	Physical manifestation	Variable	Childhood	Minor
Arachnodactyly	HP:0001166	Physical manifestation	Variable	Congenital	Minor
Renal abnormalities	HP:0000077	Clinical sign	Variable	Variable	Variable
Scoliosis	HP:0002650	Physical manifestation	Reported (case)	Childhood	Variable
Gingival hyperplasia	HP:0000212	Physical manifestation	Reported (case)	Childhood	Variable

Age of onset. Infancy to early childhood (developmental delay typically evident in the first years of life; some features such as microcephaly and dysmorphism are congenital).

Severity and progression. Highly **variable** in severity (ID mild to severe). The neurodevelopmental deficit is **static/non-progressive**; seizures are variable and may be episodic.

Quality of life impact. Lifelong intellectual disability and developmental delay substantially affect adaptive functioning, independence, communication, and education; seizures and skeletal/renal complications add further burden. No formal EQ-5D/SF-36/PROMIS data exist for this ultra-rare disorder.

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4. Genetic / Molecular Information

Causal gene. ***EMC10*** (ER membrane protein complex subunit 10), HGNC:28974, located on **chromosome 19q13.33**; reference transcript **NM_206538.3**. OMIM gene/phenotype entry **#619264**.

Pathogenic variants.

Variant (cDNA / protein)	Type	Classification	Notes
c.287delG (p.Gly96Alafs*9)	Frameshift	Pathogenic	Recurrent founder/hotspot allele; ~majority of patients; two distinct founder haplotypes (PMID: 33531666)
c.343C>T (p.Arg115Ter)	Nonsense	Pathogenic	PMID: 40150819
c.431del (p.Asp144AlafsTer3)	Frameshift	Pathogenic	PMID: 40741735
5 additional novel biallelic LoF variants	Frameshift/ nonsense/ splice	Pathogenic/ Likely pathogenic	Expanded cohort (PMID: 35684946)

Variant class and functional consequence. All disease alleles are **loss-of-function** (frameshift, nonsense, splice). The recurrent c.287delG "results in significantly reduced EMC10 RNA expression and an unstable truncated EMC10 protein" (PMID: 33531666) — i.e., LoF via nonsense-mediated decay of the transcript plus instability of any residual protein. Mechanistically this is a **loss-of-function**, not gain-of-function or dominant-negative,

mechanism in the recessive context. Note that one report raised the possibility of "dual molecular effects" on the neuropsychiatric phenotype (PMID: 40741735); nonetheless the established disease mechanism is biallelic LoF.

Allele frequency / population databases. Pathogenic *EMC10* LoF alleles are ultra-rare in gnomAD; the recurrent c.287delG is notable for arising at a **hypermutable site** rather than as a single ancient founder event.

Somatic vs germline. All variants are **germline**. No somatic association.

Modifier genes. None identified. Phenotypic variability is unexplained at the genetic-modifier level.

Epigenetic information. None specifically described for NEDDFAS.

Chromosomal abnormalities. None; disease is caused by small sequence variants, not large structural/chromosomal rearrangements. Chromosomal microarray and karyotype are typically normal (and used to exclude differentials).

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5. Environmental Information

Environmental factors: None. NEDDFAS is a purely genetic Mendelian disorder. **Lifestyle factors:** Not applicable. **Infectious agents:** Not applicable.

This section is **not applicable** beyond noting that consanguinity (a demographic/social factor) increases the probability of a child being homozygous for a rare recessive allele, but it is not an "environmental cause" of the mutation itself.

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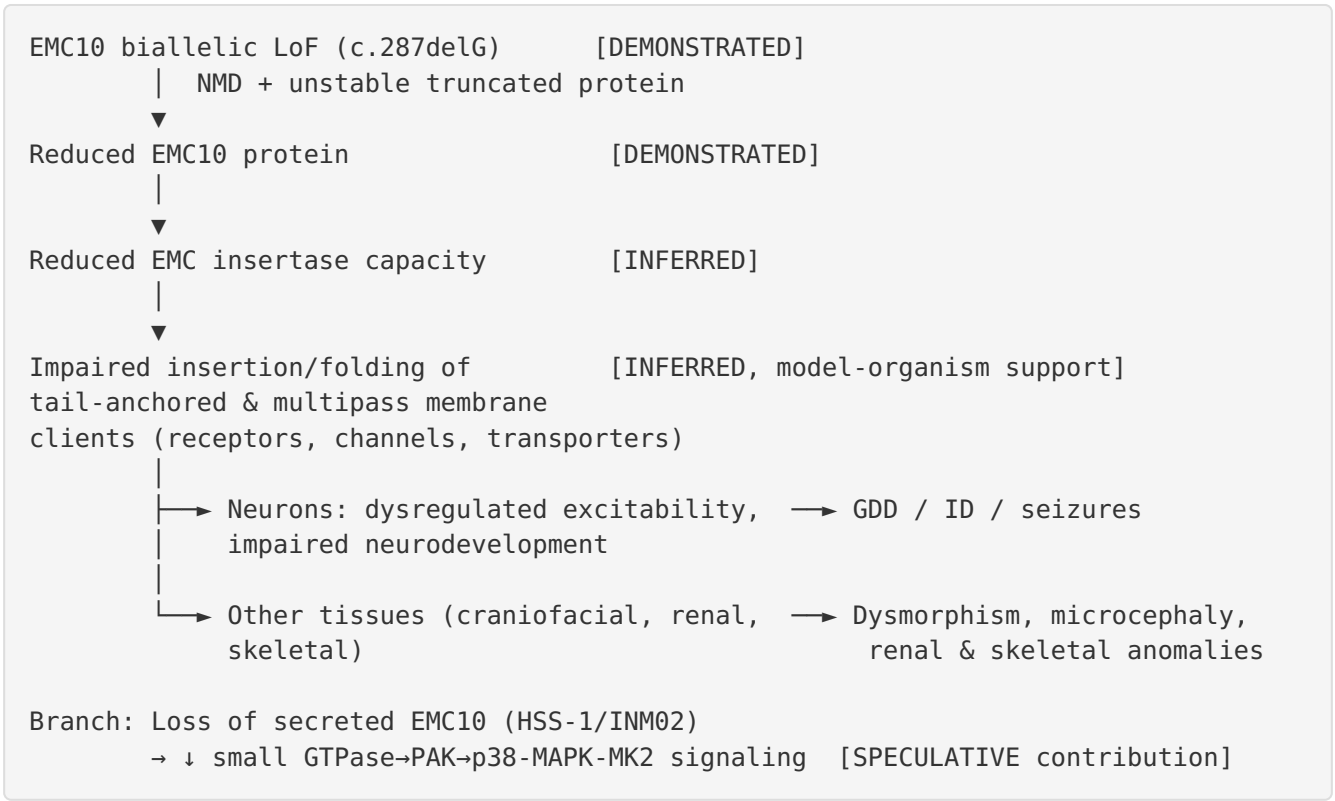
6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **biallelic loss-of-function variant in *EMC10*** (e.g., c.287delG, p.Gly96Alafs*9) is present. (Demonstrated.)*
2. The frameshift transcript undergoes **nonsense-mediated decay**, *leading to* "significantly reduced EMC10 RNA expression," and any residual protein is "an unstable truncated EMC10 protein" (PMID: 33531666). **(Demonstrated.)**

3. Loss of the EMC10 subunit *results in* **reduced functional ER membrane protein complex (EMC) insertase activity**. **(Inferred** from EMC subunit biology.)
4. Reduced EMC capacity *impairs* the **co-/post-translational insertion and folding of EMC client proteins** — tail-anchored and multipass transmembrane proteins, including neuronal receptors, ion channels, and transporters — because EMC is "responsible for the post-translational insertion of tail-anchored membrane proteins into the endoplasmic reticulum in a defined topology" (PMID: 35684946). **(Inferred** from EMC function; supported by model-organism data, e.g., PMID: 34918864.)
5. Defective biogenesis of these membrane clients *leads to* **dysregulated neuronal excitability and impaired neurodevelopment**, manifesting as GDD/ID and seizures; parallel effects in other tissues *produce* dysmorphism and renal/skeletal anomalies. **(Inferred.)**

Branch — secreted EMC10 isoform. EMC10 also has a secreted isoform (EMC10/INM02/HSS-1/SCA1) that acts as a bone-marrow-derived angiogenic growth factor: "Emc10 signaled through small GTPases, p21-activated kinase, and the p38 mitogen-activated protein kinase (MAPK)-MAPK-activated protein kinase 2 (MK2) pathway to promote actin polymerization and endothelial cell migration" (PMID: 28931551). Loss of this secreted signaling function *could* contribute to developmental/vascular phenotypes. **(Speculative.)**



Detail by category

- **Molecular pathways:** Core defect is in the **EMC-mediated membrane-protein insertion pathway** (parallel to the GET pathway) in the ER. The catalytic core belongs to the **Oxa1 insertase family** (PMID: 37417332; PMID: 33629497; PMID: 36041783). The secreted-isoform branch involves **small GTPase → PAK → p38-MAPK-MK2** signaling (PMID: 28931551).
- **Cellular processes:** Membrane-protein biogenesis, ER protein quality control/folding. In fungi, EMC loss also impaired **autophagy** via reduced Atg9 steady-state levels (PMID: 36772852) — a potential but unproven contributor in humans.
- **Protein dysfunction:** Loss of function of the EMC10 subunit; truncated protein is unstable and degraded.
- **Immune / metabolic / tissue-damage mechanisms:** No specific autoimmune, metabolic, or oxidative-stress mechanism established for NEDDFAS.
- **Cell types & biological processes (ontology suggestions):** GO:0045050 (protein insertion into ER membrane), GO:0071816 (tail-anchored membrane protein insertion into ER membrane), GO:0006612 (protein targeting to membrane), GO:0007399 (nervous system development), GO:0021954 (central nervous system neuron development). Cell types: CL:0000540 (neuron), CL:0000679 (glutamatergic neuron), CL:0000099 (interneuron).

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7. Anatomical Structures Affected

Organ level. - **Primary:** Central nervous system / brain (UBERON:0000955) — the dominant target (ID, GDD, seizures, microcephaly). - **Secondary:** Kidney (UBERON:0002113) — renal abnormalities; skeleton (UBERON:0001434) — scoliosis, cubitus valgus, arachnodactyly; craniofacial structures (UBERON:0000033, head) — dysmorphism; gingiva/oral cavity (UBERON:0001828) — gingival hyperplasia. - **Body systems:** Nervous system (primary); musculoskeletal, renal/urinary, and craniofacial systems (secondary).

Tissue and cell level. Nervous tissue (neurons, CL:0000540). Because EMC is ubiquitously required for membrane-protein biogenesis, multiple tissue types are affected, but neurons appear most clinically vulnerable. EMC10 is expressed in normal human brain by immunohistochemistry (PMID: 33531666).

Subcellular level. The **endoplasmic reticulum membrane** (GO:0005789) is the site of EMC function; the EMC is an integral ER-membrane complex. GO cellular component: GO:0072546 (EMC/ER membrane protein complex).

Localization / lateralization. Brain involvement is **bilateral/diffuse**; dysmorphic and skeletal features are generally symmetric. No consistent lateralization.

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8. Temporal Development

Onset. Congenital to early childhood. Developmental delay is apparent in infancy; microcephaly and dysmorphism may be present at birth. Onset is **insidious/chronic** rather than acute.

Progression. The intellectual disability is **static (non-progressive)** — a stable neurodevelopmental deficit rather than a neurodegenerative course. This contrasts with EMC1-related disease, which can feature cerebellar atrophy/degeneration (PMID: 26942288). Seizures may be episodic. Disease duration is **lifelong/chronic**.

Patterns / critical periods. No spontaneous remission of the core neurodevelopmental phenotype. The key window for intervention is **early childhood** (early developmental intervention, seizure control), consistent with general neurodevelopmental-disorder principles.

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9. Inheritance and Population

Epidemiology. Ultra-rare. No formal prevalence or incidence estimates exist. Approximately **25–30 reported patients across fewer than 15 families** at the time of research. The disorder is enriched in **consanguineous populations** and international in distribution (reported in Middle Eastern, South Asian, and other consanguineous cohorts).

Inheritance. Autosomal recessive. Affected individuals are homozygous (or compound heterozygous) for biallelic LoF *EMC10* variants in consanguineous families (PMID: 33531666; PMID: 35684946). **No monoallelic (dominant) EMC10 disease has been established** — a notable contrast with *EMC1*, where both monoallelic and biallelic variants cause disease (PMID: 26942288).

Penetrance / expressivity. Appears **fully penetrant** in biallelic LoF individuals, with **variable expressivity** (ID severity mild to severe; variable seizures and dysmorphism).

Founder effect / hypermutable site. The recurrent c.287delG "arose on two founder haplotypes" and, remarkably, "is likely the result of a hypermutable site and arose on distinct founder haplotypes" (PMID: 33531666) — i.e., recurrence is driven by a mutational hotspot rather than a single shared ancestor.

Anticipation / germline mosaicism. No genetic anticipation (not a repeat-expansion disorder). Germline mosaicism not specifically reported.

Consanguinity. A major contributing factor, as expected for a rare recessive disorder.

Carrier frequency. Not established; ultra-rare alleles make population carrier frequency very low outside affected families.

Demographics. No strong sex bias established (autosomal recessive; males and females affected). Age distribution reflects pediatric diagnosis.

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10. Diagnostics

Genetic testing (the diagnostic cornerstone). All reported cases were identified by **whole-exome sequencing (WES) or whole-genome sequencing (WGS)** with **Sanger confirmation and familial segregation analysis**. A representative diagnostic pathway: "Initial genetic testing included karyotype analysis, FMR1 CGG repeat analysis, and chromosomal microarray analysis. Subsequently, whole-exome sequencing (WES) was performed, and Sanger sequencing was used to confirm the identified variant and conduct familial segregation analysis" (PMID: 40741735). The value of reanalysis is emphasized: Forest-St-Onge et al. "underscore the importance of exome reanalysis in clinical practice" (PMID: 40150819).

- **WES / WGS:** High diagnostic utility; the primary method by which the disorder is identified.
- **Gene panels / single-gene testing:** ID/epilepsy gene panels or *EMC10* single-gene testing can detect the recurrent c.287delG, especially in consanguineous families or those of shared ancestry.
- **Chromosomal microarray, karyotype, FMR1 (Fragile X) testing:** Typically **normal**; used to exclude common differentials before/alongside sequencing.
- **Mitochondrial DNA / repeat-expansion testing:** Not indicated (not the disease mechanism).

Clinical and imaging workup. Brain **MRI** and **EEG** (for seizure characterization) support the clinical assessment. Renal imaging/ultrasound where renal abnormalities are suspected.

Biomarkers / laboratory tests. No specific biochemical biomarker exists; diagnosis is genetic.

Clinical criteria / differential diagnosis. No formal consensus diagnostic criteria. Key **differential diagnoses** include **EMC1-related neurodevelopmental disorder** (overlapping features; "an overlap between the clinical features of EMC10- and EMC1-related disease," PMID: 35684946) and other syndromic autosomal recessive intellectual-disability disorders.

Screening. In families with a known *EMC10* variant, **cascade carrier testing** is appropriate. No population-based newborn screening exists for this ultra-rare disorder.

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11. Outcome / Prognosis

Survival / mortality. No disease-specific mortality data and **no evidence of reduced survival** reported. Life expectancy appears not markedly shortened based on available (limited) reports.

Morbidity and function. The dominant burden is **lifelong intellectual disability and developmental delay** of variable severity, with associated impacts on education, communication, independence, and adaptive functioning. Seizures, scoliosis, renal anomalies, and dental issues add morbidity. No formal disability/QoL instrument data exist.

Disease course. Static (non-progressive) neurodevelopmental deficit; chronic and lifelong. Complications are managed symptomatically.

Prognostic factors. Severity of baseline intellectual disability and seizure control are the main determinants of functional outcome. No validated molecular prognostic biomarkers; given the small cohort, genotype–phenotype correlation (beyond "all LoF") is not well established.

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12. Treatment

There is no disease-specific or disease-modifying therapy. Management is **supportive and symptomatic**, following standard practice for syndromic neurodevelopmental disorders.

Domain	Intervention	NCIT suggestion
Epilepsy/seizures	Antiseizure medications (chosen by seizure type)	NCIT:C264 (Anticonvulsant Agent)
Developmental delay	Early intervention, special education	NCIT:C15215 (Rehabilitation Therapy)
Motor/functional	Physical and occupational therapy	NCIT:C15351 (Physical Therapy); NCIT:C15402 (Occupational Therapy)
Communication	Speech/language therapy	NCIT (Speech Therapy)
Skeletal (scoliosis)	Orthopedic management/ monitoring	NCIT:C157866 (Orthopedic Procedure)
Dental (gingival hyperplasia)	Dental care	—
Renal abnormalities	Nephrology monitoring/ management	—

- **Pharmacogenomics:** No disease-specific pharmacogenomic guidance; standard antiseizure medication selection applies.
- **Advanced therapeutics (gene/cell/RNA therapy, targeted/immunotherapy):** None available or in trials for NEDDFAS.
- **Experimental treatments / clinical trials:** None identified (no NCT-registered NEDDFAS-specific trials).
- **Treatment outcomes / adverse events:** Not formally studied for this disorder; outcomes relate to symptomatic control.

13. Prevention

NEDDFAS cannot be prevented at the individual (mutation) level; prevention focuses on **reproductive genetic risk management**.

- **Primary prevention:** Not applicable (genetic disorder).

- **Genetic counseling:** The cornerstone of prevention for at-risk families, particularly **consanguineous couples** and known carriers. Counseling covers the 25% recurrence risk per pregnancy for carrier couples.
- **Carrier / cascade testing:** Available once the familial variant is known.
- **Prenatal diagnosis and preimplantation genetic testing (PGT):** Available for families with a known *EMC10* variant.
- **Immunization / behavioral / public-health / environmental interventions:** Not applicable.
- **Tertiary prevention:** Prevent complications via seizure control, orthopedic surveillance for scoliosis, renal monitoring, and dental care.

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14. Other Species / Natural Disease

- **Taxonomy / naturally occurring disease:** No naturally occurring *EMC10*-related disease is catalogued in companion animals or wildlife (no OMIA entry identified). **Not applicable** beyond experimental models.
- **Orthologous genes:** *EMC10* is conserved; mouse *Emc10* (used in the angiogenesis/myocardial-infarction work, PMID: 28931551). The **EMC complex itself is deeply conserved** from yeast and fungi (e.g., *Magnaporthe oryzae* MoEmc2/MoEmc5, PMID: 36772852) through *Drosophila* (PMID: 34918864) to humans, underscoring strong **evolutionary conservation of the disease-relevant mechanism** (membrane-protein insertion).
- **Zoonotic potential / transmission:** Not applicable (non-communicable genetic disorder).

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15. Model Organisms

No dedicated published NEDDFAS (*EMC10*-specific) disease model recapitulating the full human syndrome was identified at the time of research. However, the **EMC complex** has been extensively studied across model systems, providing strong mechanistic support:

System	Finding	Relevance	Reference
Mouse (*Emc10*)	Secreted Emc10 is a bone-marrow-derived angiogenic growth factor promoting tissue repair after myocardial infarction	Documents the secreted-isoform signaling branch	PMID: 28931551
Drosophila	EMC required for biogenesis of Xport-A, Rhodopsin-1, and TRP channel; EMC needed for TMD membrane insertion	Demonstrates EMC's essential role in membrane-protein (incl. ion channel) biogenesis	PMID: 34918864
Fungus (*M. oryzae*)	EMC loss reduces membrane-protein steady-state levels, impairs autophagy, cell-wall integrity, pathogenicity	Shows conserved EMC function and client-protein dependence	PMID: 36772852
Yeast→human (structural)	Cryo-EM structures reveal conserved EMC architecture and substrate TMD-binding pocket	Structural basis of insertase mechanism	PMID: 33629497; PMID: 35850079

Applications and gaps. These models establish *what EMC does* but do not yet model the *EMC10-deficient neurodevelopmental phenotype*. A faithful mammalian neuronal or whole-animal *Emc10*-LoF model (knockout/knock-in of the patient allele) that reproduces developmental delay/seizures is a key unmet need (see Follow-up).

Mechanistic Model / Interpretation

NEDDFAS is best understood as a **membrane-protein biogenesis disorder of the endoplasmic reticulum**. The unifying model: biallelic *EMC10* LoF → reduced/degraded EMC10 protein → diminished EMC insertase capacity → impaired insertion and folding of tail-anchored and multipass membrane clients → cell-biological dysfunction most clinically evident in neurons (ID, GDD, seizures) with additional craniofacial, renal, and skeletal manifestations. Steps 1–2 are **demonstrated** in patient material (PMID: 33531666); steps 3–5 are **inferred** from

robust, cross-species EMC biology (PMID: 35684946, PMID: 34918864, PMID: 33629497, PMID: 36041783). A secondary, speculative branch involves loss of the **secreted EMC10 isoform** and its small-GTPase→PAK→p38-MAPK-MK2 angiogenic signaling (PMID: 28931551).

The **close parallel to EMC1-related disease** (PMID: 26942288) is informative: both are EMC-subunit disorders causing syndromic ID, but EMC1 disease prominently features cerebellar atrophy/degeneration and can be dominant or recessive, whereas EMC10 disease is recessive and static. This suggests that **different EMC subunits, when disrupted, produce overlapping but distinguishable neurodevelopmental syndromes**, reflecting partially distinct client repertoires or dosage sensitivities.

Evidence Base

PMID	Paper (abbrev.)	Role in this report
33531666	Recurrent homozygous *EMC10* frameshift → syndrome (Shao 2021)	**Foundational.** Defines disease, core phenotype, recurrent c.287delG, two founder haplotypes/ hypermutable site, LoF molecular consequence
35684946	Biallelic *EMC10* LoF → mild-to-severe ID (Kaiyrzhanov/ Maroofian 2022)	**Confirmatory/expansion.** 10 more patients, 5 new LoF variants; EMC10 function definition; EMC1 overlap
40150819	Gingival hyperplasia & scoliosis as added features (Forest-St-Onge 2025)	Multisystem phenotype (microcephaly, renal); c.343C>T; exome reanalysis
40741735	*EMC10* dual molecular effects (Bolat 2025)	Diagnostic pathway; c.431del; possible dual molecular effect
28931551	Secreted EMC10 angiogenic factor (Reboll 2017)	Secreted-isoform signaling branch (GTPase/PAK/p38-MAPK-MK2)
26942288	*EMC1* variants in NDD (Harel 2016)	Differential diagnosis; EMC-subunit disease comparison
37417332	Membrane insertases at a glance	EMC as Oxa1-family insertase
33629497	Cryo-EM of EMC	Structural basis of insertase; substrate TMD pocket
36041783	GET vs EMC TA-insertion pathways	Pathway context
35850079	Cryo-EM of TA biogenesis	Conserved insertase core
34918864	EMC & Xport-A/Rh1/TRP (Drosophila)	Model-organism evidence for client biogenesis incl. channels
36772852	Fungal EMC & membrane proteins/autophagy	Conserved function; autophagy link

Consistency. All disease-focused reports agree on autosomal recessive biallelic LoF of *EMC10* as the cause and converge on the GDD/ID + variable seizures + dysmorphism core, with progressive broadening of the phenotype (microcephaly, renal, skeletal, dental) as more cases accrue. The mechanistic literature (structural, model-organism) consistently supports the EMC insertase role, though direct demonstration in human NEDDFAS neurons is still lacking.

Limitations and Knowledge Gaps

1. **Small evidence base.** ~25–30 patients across <15 families; frequencies are qualitative ("variable"), and robust genotype–phenotype correlations and prevalence/incidence are unavailable.
2. **Inferred downstream mechanism.** Steps beyond reduced EMC10 protein (insertase-capacity loss, specific affected clients, neuronal consequences) are **inferred from EMC biology**, not directly demonstrated in NEDDFAS patient neurons. The specific client proteins whose mislocalization drives the neurodevelopmental phenotype are unknown.
3. **No animal model of the syndrome.** Existing *Emc10*/EMC models address function or angiogenesis, not the human neurodevelopmental disorder.
4. **Unexplained variability.** No modifier genes or epigenetic factors identified to explain the mild-to-severe range of ID.
5. **Secreted-isoform branch speculative.** The contribution (if any) of lost secreted EMC10 signaling to the human phenotype is untested.
6. **"Dual molecular effects"** (PMID: 40741735) is intriguing but not mechanistically resolved.
7. **No therapeutics.** No disease-modifying treatment or trials.

Proposed Follow-up Experiments / Actions

1. **Generate a patient-allele mouse model** (knock-in *Emc10* p.Gly96Alafs*9 or conditional neuronal knockout) and assess developmental, behavioral, EEG/seizure, and craniofacial/renal phenotypes to test the inferred causal chain.
2. **Patient-derived iPSC neurons/organoids:** quantify EMC assembly/stability and perform surfaceome/membrane-proteomics (e.g., quantitative MS) to identify which EMC client proteins (receptors, ion channels, transporters) are depleted — directly testing step 4.

3. **Electrophysiology** in patient neurons/organoids to connect client-protein deficits to the inferred excitability changes underlying seizures.
4. **Natural-history registry** for *EMC10* patients to establish prevalence, age-dependent features, seizure semiology, QoL, and genotype–phenotype correlation.
5. **Test the secreted-isoform branch** by measuring circulating EMC10 and vascular/angiogenic readouts in patients vs controls.
6. **Systematic EMC10 vs EMC1 comparison** (and other EMC subunits) to map subunit-specific client dependencies that explain the distinct clinical syndromes.
7. **Variant curation:** submit all reported alleles to ClinVar with ACMG classifications and establish an *EMC10* gene–disease ClinGen curation to standardize diagnosis.

Report compiled from 6 confirmed findings and 12 reviewed papers over a 5-iteration investigation. Evidence source types: human clinical (case series/reports), model organism (mouse, Drosophila, fungus), and structural/computational (cryo-EM). All mechanistic steps are flagged as demonstrated, inferred, or speculative.
